

Airline Kiosk & Baggage Check — Pre Authentication, Context Collection & Identity Normalization

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Executive Summary

In the Enterprise-Trust Architecture model, the airport flow begins long before TSA, and Section 4.1.1 marks the true architectural starting point — the moment a request first interacts with the enterprise. This stage is not authentication and not authorization; it is context collection: identity lookup, attribute changes, device posture signals, privilege updates, and environmental inputs that shape the request before it reaches the first real enforcement point, the Identity Gateway (TSA).

But this early phase performs a second, equally critical function: Identity Normalization, the creation of the enterprise's boarding pass. Whatever identity the user brings — Entra ID, federation, workload identity, anything — is converted into the single token format the enterprise understands through the Internal STS, the service most organizations rely on but rarely describe clearly. The Internal STS acts as the enterprise's token normalization engine — the service that converts whatever upstream identity token the user brings into the single, unified token format the enterprise trusts.

Note: The Internal STS is used for both guest and workforce identities, but the flows are not identical. This chapter focuses on the public-side path; the workforce path appears later in 4.5.

This chapter expands Section 4.1.1 into a full, standalone narrative that recaps the broader Enterprise-Trust Architecture, introduces the Trust Boundary Compass™, and dives deep into the mechanics of Pre-Authentication: what it is, what it is not, and why confusing this preparatory stage with trust inevitably leads to architectural breakdown.

4.1.1 Context: Where We Are on the Enterprise-Trust Flow

In the Enterprise-Trust Architecture flow, the airport metaphor frames identity as the foundational security boundary, and at this stage the traveler is still on the public side of the terminal.

4.1.1.1 External Network → Arriving at the Airport

You begin as an unknown identity — anyone can walk into the non-secure area of an airport, the enterprise equivalent of an unauthenticated external network. Maybe you're flying, maybe you're picking someone up, maybe you're just grabbing lunch. No trust decision has been made yet.

Before you even reach the airline kiosk, you pass through the airport's outer security presence — uniformed officers, cameras, and basic screening designed to keep obvious threats out of the building. This is not identity verification and not part of the Enterprise-Trust flow. It maps to the enterprise's Threat PEP layer — Akamai, Cloudflare WAF, Fastly — which filters malicious traffic before the request reaches the identity pipeline.

It keeps the lobby safe, but it does not grant trust, issue tokens, or activate the Compass.

You're simply another face in the lobby, walking past the donut shop and heading toward the miles of roped-off tunnels that eventually lead to TSA. The identity flow begins only when you step up to the kiosk.

4.1.1.2 Airline Kiosk & Baggage Check → Pre-Authentication & Context Collection

The first meaningful interaction occurs when you approach the airline kiosk or counter. That moment signals intent: you want access to the secure side of the airport.

This is where Pre-Authentication and context collection begin — your identity is looked up, attributes are refreshed, device posture is gathered, and privileges are checked, all before the request moves toward the Identity Gateway (TSA).

This is the transition point where the enterprise finally has something to work with: not trust, but context.

4.1.1.3 The Trust Boundary Compass™

Before we go any deeper into Pre-Authentication, we have to introduce the Trust Boundary Compass — the model that evaluates every request as it moves through the Enterprise-Trust environment.

The Compass defines four pillars:

- Identity — Who or what is making the request Human identities, workload identities, service principals, managed identities, federated identities, and the claims, attributes, and privileges attached to them.
- Device — What the request is coming from Compliance, attestation, OS version, risk posture, token binding, configuration, and health.

- Network — Where the request is coming from Network path, segmentation, service mesh identity, east-west authentication, and boundary transitions. Enterprise-Trust rejects the old “trusted network” model entirely.
- Workload — What the request is trying to do Workload identity, API behavior, service-to-service calls, token exchange, multi-hop identity propagation, and workload posture or drift.

Pre-Authentication is the first moment the Compass activates, but none of these pillars grant trust yet — they only collect context so the Identity Gateway (TSA) can make an informed decision.

4.1.1.4 Why the Compass Matters Here

The Trust Boundary Compass appears before the enforcement stage for a reason: you need the map before you start the trip.

Pre-Authentication — the airline kiosk moment — is where the Compass first wakes up and starts doing its thing:

- Identity asks whether you even exist
- Device wants to know what you’re using
- Network checks where you wandered in from
- Workload wants to know what you’re trying to do

But — and this is the part people love to get wrong — none of these pillars grant trust. They’re just collecting context so the first real enforcement point, the Identity Gateway (TSA), can make an informed decision.

In the next section, you’ll see the Compass snap awake and start judging your shoes, your bags, and your life choices — usually right before it tells you to throw away the donut you just bought.

Placing the Compass here gives the reader the evaluation model before they enter the Pre-Authentication zone, so they understand exactly what’s happening when the Compass snaps awake and starts evaluating you for real.

4.1.1.6 Airline Kiosk & Baggage Check → Pre-Authentication & Context Collection

Pre-Authentication — the airline kiosk zone of the Enterprise-Trust Architecture flow — is often misunderstood, so it’s worth stating plainly: this stage is preparation, not permission.

When you walk up to TSA, you don't hand them your online reservation, your credit card, or your Entra ID login. You present a boarding pass — either the printed version generated by the kiosk or the mobile version already bound to your phone.

When a request steps up to the “kiosk,” the enterprise isn't trusting it; it's collecting the raw materials needed for someone or something else to make the real call. Identity is looked up, device posture is evaluated, attributes and privileges are refreshed, and all of that context is packaged into the enterprise's version of a boarding pass: the normalized identity token.

That token is the format every downstream system understands, and it's generated here — not at the Identity Gateway, and not during authentication proper.

The kiosk zone's entire job is to gather context, normalize identity, and hand the request off to the first true enforcement point with everything it needs. No access is granted. No trust is implied. This is simply the prep work before the real decision happens.

Takeaway: Pre-Authentication does not grant access. It gathers the identity and device context required for the Identity Gateway to make an informed decision — nothing more.

4.1.1.7 The Kiosk → Output Medium (Not the STS)

TSA never evaluates your upstream identity provider. TSA evaluates the token the STS issued.

The kiosk is not the STS, and it's definitely not the identity authority — it's just the output medium. All the kiosk does is print the paper boarding pass the enterprise already issued; the same way the airline website or mobile app simply delivers the digital version to your phone. Both are external delivery mechanisms, nothing more.

The actual identity authority is the IdP — Entra ID in our world — the system that stores your identity, validates your credentials, enforces MFA, maintains your attributes, and determines your account state.

The Internal STS, sitting safely inside your trust boundary, is the engine that takes that upstream identity and transforms it into the single, standardized enterprise token. It also bridges protocol gaps — taking modern IdP tokens (OIDC/SAML) and translating them into the formats required by legacy and on-prem applications. It validates the identity the IdP proved, applies policy, normalizes claims, signs the token, enforces schema and lifetime, manages trust anchors, and handles legacy compatibility.

Some enterprises use:

- PingFederate
- ForgeRock
- Keycloak
- A home-grown STS in a container
- A legacy STS with token transformation plugins

It doesn't matter which one you use.

What matters is: The STS must be inside your trust boundary.

Because only you control:

- signing keys
- claim schema
- token lifetime
- trust anchors
- backward compatibility
- legacy integration

The Internal STS gives you identity sovereignty.

4.1.1.7a So What Does the STS Actually Do?

The STS doesn't decide who you are; it converts who you are into the one boarding pass format the entire airport understands.

That's why you end up with two boarding passes — one printed, one mobile — each representing a different binding model.

Wait — two boarding passes? Yes.

- The printed pass is identity-only: standardized, TSA-trusted, physical, and not device-bound.
- The mobile pass is identity-plus-device: cryptographically bound to your phone, non-transferable, possession-proof, and dynamically refreshed.

Both represent identity because you proved who you are to get them, but only one carries device binding.

This distinction is the backbone of Enterprise-Trust Architecture, and it's why token normalization is mandatory.

Your environment contains many identity formats and protocols:

- cloud services (OIDC, SAML)
- on-prem services
- legacy AD and LDAP
- SOAP and WS-Trust services
- mainframe endpoints
- API gateways
- service mesh identities
- microservices and containers

The Internal STS bridges these worlds — taking modern IdP tokens (OIDC/SAML) and transforming them into the single enterprise token format required by downstream systems so the enterprise can function as a unified system — the “one boarding pass for the whole airport” model.

Note on the Metaphor

The airport example uses two boarding passes — a printed pass and a mobile pass — to show the difference between identity alone and identity bound to a device. This helps illustrate why device posture and token binding matter.

However, in the enterprise, you do not receive two tokens. You receive one: the normalized enterprise token issued by the Internal STS. That token contains both categories of information — identity attributes from the IdP and device binding signals collected during Pre-Authentication.

The metaphor separates them so the reader can see the distinction. The architecture merges them so the enterprise can operate as a unified system.

4.1.1.8 Identity Lookup at the Counter → “Do You Even Exist?”

When you walk up to the airline kiosk, the very first thing it does is look you up. Before it prints anything, updates anything, or offers you anything, the system checks your name, reservation, flight, seat, baggage allowance, and status.

This is the airline verifying that you exist in its system. If you don’t? Nothing else proceeds. No bags. No upgrades. No boarding pass. You’re done.

Enterprise-Trust Architecture works exactly the same way: no identity record = no flow. The environment must confirm that the identity exists before any other signal — device, network, workload — matters.

This is the IdP lookup, the foundational “Do you exist?” moment that everything else depends on.

Entra ID Example

When a user enters their UPN at the kiosk, Entra ID performs a directory lookup to confirm the identity exists and retrieve baseline attributes such as account status, tenant membership, and assigned licenses. No token is issued and no trust is granted — this is simply the identity equivalent of the airline verifying your reservation. If the identity doesn’t exist, the flow ends immediately.

Compass Update

- Identity: Directory object found; baseline attributes loaded
- Device: Unknown
- Network: Source IP + geolocation captured
- Workload: Interactive login intent detected

Identity exists — but it is not trusted.

4.1.1.8a Human vs. Non-Human Entry Paths

Humans can exist in the “public area” without identity. Non-human identities cannot.

APIs, workloads, and AI agents never casually appear inside an enterprise. They don’t wander the lobby. They don’t browse the gift shop.

They only “arrive” when they present:

- a client credential
- a workload identity

- a managed identity
- a service principal
- or an agent persona

Where humans begin with zero context, non-human identities begin with declared intent.

Both follow the same architectural stages, but their entry conditions differ:

Humans: Public area → kiosk → TSA → secure zone

Non-humans: Identity assertion → Pre-Authentication → Gateway → secure zone

4.1.1.9 Checking Bags → Attribute & Context Updates

Checking a bag doesn't authenticate you — it simply updates your attributes. Your baggage count changes, your weight changes (not on the good side), your fees change, and your reservation metadata updates.

Nothing about that interaction makes you more trusted; it just enriches the airline's understanding of your state.

Enterprise-Trust Architecture treats these signals the same way: a new device, new claims, new workload metadata, or new entitlements are telemetry, not trust. They update the identity object, but they don't move you any closer to the secure side of the airport.

Entra ID Example

When a device registers with Entra ID or updates its compliance posture, the directory records new attributes such as device ID, compliance state, OS version, and attestation status. These updates do not authenticate the user — they simply enrich the identity object with fresh context.

Compass Update

- Identity: Attributes enriched
- Device: Compliance posture collected
- Network: Logged
- Workload: Unchanged

Context increases — trust does not.

4.1.1.10 Seat Upgrades → Attribute Changes That Affect Later Access

Upgrading your seat — whether at home, on your phone, or at the counter — does not bypass TSA. You don't get to stroll past the line because you're in 2A instead of 32B.

A seat upgrade simply changes what you're entitled to after you authenticate.

Enterprise-Trust Architecture handles privilege changes the same way: PIM elevation, new RBAC roles, updated claims, or new privileges modify what you can do later — not what you can bypass now.

Entra ID Example

When a user activates a PIM role, Entra ID updates the identity's effective privileges and issues time-bound role metadata. This does not bypass Conditional Access or device compliance checks — it only changes what the user may be entitled to after authentication.

Compass Update

- Identity: Privilege metadata updated
- Device: Unchanged
- Network: Logged
- Workload: Unchanged

Attributes change — trust does not.

4.1.1.11 Adding a Second Device → Context Change, Not Trust

If work gives you a laptop and later adds a tablet, your identity now has two devices instead of one. That's a state change, not trust.

Enterprise-Trust Architecture records each device's ID, compliance posture, attestation state, OS version, and risk baseline.

More devices simply mean more context for the system to evaluate later — not a shortcut through the checkpoint.

Entra ID Example

When a user enrolls a second device into Entra ID, the directory simply adds another device object linked to the identity. Each device has its own compliance posture, risk score, and attestation state.

Compass Update

- Identity: Multiple device objects linked
- Device: New device posture collected
- Network: Logged
- Workload: Unchanged

More devices = more context, not more trust.

4.1.1.12 Buying a Donut Before TSA → Context That Doesn't Cross the Boundary

You can buy a donut right in front of TSA. You can hold it, admire it, cherish it, and enjoy every warm, sugary moment.

But the second you step up to the checkpoint, TSA makes you throw it away.

This isn't personal. It's not TSA "taking your donut." It's boundary policy enforcement.

They don't care that you bought it inside the airport, that you paid for it 30 seconds ago, or that it looks harmless. The rule isn't about you — it's about protecting the system.

Enterprise-Trust Architecture works the same way: device posture, token binding, risk score, compliance, and claims are evaluated at the boundary, not before it.

And yes, sometimes it feels obsessive. But it only takes one donut bomb to make things go sideways.

Entra ID Example

A user may browse internal public-facing resources or interact with pre-auth endpoints before authentication, but none of that activity carries forward into the trust decision.

Compass Update

- Identity: No change
- Device: No change
- Network: Logged
- Workload: Pre-auth activity ignored

Pre-auth behavior is not trust input.

4.1.1.13 Pre-Authentication Summary (Enterprise-Trust Model)

Before TSA, nothing is trust. Before TSA, nothing is access. Before TSA, nothing crosses the boundary.

Everything that happens before the Identity Gateway (TSA) is pure context collection — the system learning about the request, not granting anything to the request.

The Pre-Authentication phase gathers:

- identity lookup — “Who are you supposed to be?”
- attribute updates — “What changed about you since last time?”
- device changes — “Is the device still healthy, compliant, and recognizable?”
- entitlement changes — “What are you allowed to do now versus yesterday?”
- environmental signals — “Where are you coming from, and what’s happening around you?”
- behavioral patterns — “Is this normal for you, or is something off?”

All of this enriches the request. None of this grants access. None of this is trust. None of this crosses the boundary.

Entra ID Example

Entra ID collects identity attributes, device posture, risk indicators, and entitlement metadata during Pre-Authentication, but none of these signals grant access on their own. Entra ID assembles the full Compass snapshot — “Preparing for TSA.”

Compass Update

- Identity: Fully enriched
- Device: Posture collected
- Network: Path recorded
- Workload: Intent identified

Compass Update has been completed. All the pillars have been identified as needed. The next step can be taken.

4.1.2 End-to-End Takeaway

The Identity Gateway (TSA) is the first real enforcement point. Pre-Authentication prepares the request; the Gateway decides what actually gets through.

End of Section 4.1.1 — Heading to TSA

Alright, that wraps 4.1.1. Grab your shoes, finish your donut, and meet me in 4.2 — TSA is where the real trust decision happens, and the Identity Gateway doesn't negotiate with pastries.

Sources (Pre-Authentication & Context Collection)

- **NIST SP 800-63-3 — Digital Identity Guidelines**
- **NIST SP 800-207 — Zero Trust Architecture**
- **NIST SP 800-204B — Attribute-Based Access Control**
- **Microsoft Entra ID Conditional Access & Device Compliance**
- **Azure AD PIM (Privileged Identity Management)**
- **OpenID Connect Core**
- **TSA Security Screening Procedures**
- **OWASP API Security Top 10**